MILITARY SPECIFICATION

OSCILLOGRAPH, TYPE V

(For Torpedo Mark 46 Mod 0)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the detail requirements for the design and procurement of Oscillograph, Type V, BUWEPS Drawing 2090290.

2. APPLICABLE DOCUMENTS

2.1 The following specification of the issue in effect on the date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-23560(Wep)	Oscillographs, General Specification for

(Copies of specifications required by contractors in conjunction with specific procurement functions should be ordered from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with Specification MIL-O-23560(Wep) and as specified herein. In the event of conflict between the requirements of Specification MIL-O-23560(Wep) and this detail specification, the requirements herein shall govern.

3.2 Design and construction shall be in accordance with BUWEPS Drawing 2090290 and Specification MIL-O-23560(Wep).

FSC 1355

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3.2.1 The viewing screen shall be marked with a scale graduated in millimeters (mm), with each tenth graduation from 0 to 90 mm marked. The screen shall be positioned with the zero calibration at the end of the magnetic structure containing galvanometer element 14. Graduations shall be accurate within plus or minus 0.005 of an inch.

3.2.2 The galvanometers shall be adjusted so that the neutral positions (no signal applied) of the traces are evenly spaced between the 10 mm and 80 mm graduations, with galvanometer No. 1 falling on the 80 mm graduation, galvanometer No. 2 falling on the 75 mm graduation, and so on. The traces shall appear at the required graduations within plus or minus 0.5 millimeter.

3.2.3 The galvanometers shall be in accordance with BUWEPS Drawing 1626671-6, and the galvanometer channels shall meet the requirements of Tables I and II.

3.2.4 Motor and lamp circuits. The motor and lamp circuits shall be in accordance with BUWEPS Drawing 2090290 and shall operate at 40 volts dc plus or minus 5 volts. The total current required to drive the motor and lamp shall be less than 2.5 amperes. The lamp and motor shall be electrically insulated one from the other. The lamp may be rated at other than 6 volts (GE Number 1926 or equivalent).

3.2.5 Static reference marker. A mechanical attachment on the magnetic structure near channel number 14 shall provide a static reference trace on the recording film. Reference trace shall be adjusted to appear at 15 millimeters on the viewing screen or recording film.

3.2.6 Magnetic interference. The oscillograph, when mounted and operated in the test configuration shown in Figure 1, and in accordance with calibration and test procedures of 4.1.2.1, shall not produce a voltage, transient or otherwise, greater than the peak voltage observed in 4.1.2.1.1.3 of the calibration procedure. The output must be viewed on the oscilloscope during the entire time required to expose a 40-foot roll of fiml in a one-inch-per-second magazine (approximately 8 minutes).

3.2.7 Weight. The weight of the oscillograph shall not exceed 9 pounds.

3.2.8 Film driving mechanism. Only one film speed need be provided, which shall be 1 inch per second plus or minus 5 percent. Requirement 3.4.4 of MIL-O-23560(Wep) applies otherwise.

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Channel Number	Pin Connection on P4804 Polarity		Sensitivity		Direction of Deflection from Static Reference*
	Positive	Negative	Input to P4804 Pins (ma)	Deflection at Viewing Screen (mm)	
1	n	p	0.03 ± 2%	8.9 mm ± 12%	Away from static reference (Up)
2	k	m	"	"	" " " "
3	i	j	"	"	" " " "
4	g	h	"	"	" " " "
5	e	f	"	"	" " " "
6	c	d	"	"	" " " "
7	a	b	"	"	" " " "
8	Y	Z	"	"	" " " "
9	W	X	"	"	" " " "
10	U	V	"	"	" " " "
11	S	T	"	"	" " " "
12	P	R	"	"	" " " "
13	M	N	"	"	" " " "
14	K	L	"	"	" " " "
15	Static Reference (Located at 15 millimeters)				

*When film is read emulsion side up, and the starting end of film is at the observer's left.

Table I

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Channel Number	Pin Connection on P4804 Positive	Negative	Voltage (rms) Millivolts	Natural Freq. (cps)	Frequency Range (cps)	Frequency Response Flat Within @ 100°F	*@ 0°F	*±160°F
1	n	p	1.0 ± 0.1	200	0 - 120	± 7.5%	+7.5% - 15%	+ 15% -7.5%
2	k	m	"	"	"	"	"	"
3	i	j	"	"	"	"	"	"
4	g	h	"	"	"	"	"	"
5	e	f	"	"	"	"	"	"
6	c	d	"	"	"	"	"	"
7	a	b	"	"	"	"	"	"
8	Y	Z	"	"	"	"	"	"
9	W	X	"	"	"	"	"	"
10	U	V	"	"	"	"	"	"
11	S	T	"	"	"	"	"	"
12	P	R	"	"	"	"	"	"
13	M	N	"	"	"	"	"	"
14	K	L	"	"	"	"	"	"
15	Static Reference (Located at 15 millimeters)							

*Required during low and high temperature tests. (Para. 4.6.4 of MIL-0-23560(Wep).

Table II

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3.3 Environmental requirements. Unless otherwise specified, the oscillograph shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed, at any temperature in the applicable range, to the environments under all unit conditions specified in this section.

3.3.1 Vibration. The oscillograph shall be capable of withstanding the vibrations in Table III.

Table III

Vibration Schedule

Unit Condition	Double Amplitude (in.) minimum	Acceleration (g) minimum	Frequency (cps)
Nonoperating	0.75	--	5-6
	--	1.3	6-16
	0.1	--	16-24
	--	3.0	24-40
	0.036	--	40-74
	--	10.0	74-500

3.3.2 Shock. The oscillograph shall be capable of withstanding shock specified in Table IV.

Table IV

Shock Schedule

Unit Condition	Peak Magnitude (g's)	Minimum Duration (ms)	Acceleration Direction
Nonoperating	60	8.0	Any
Ready, nonoperating	45	20.0	Fwd
	85	0.5	Any
	300	0.5	Aft

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3.3.3 Acceleration. The oscillograph shall be capable of withstanding the accelerations specified in Table V.

Table V

Acceleration Schedule

Condition	Magnitude (g)	Acceleration Direction
Nonoperating	25	Longitudinal Forward
Ready, non-operating	35	Longitudinal Aft
Operating	15	Each direction all axis

3.3.4 Temperature. The oscillograph shall be capable of conforming to the requirements of this specification at any temperature within the limits of Table VI, in the unit conditions specified in Table VI.

Table VI

Temperature Schedule

Condition	Temperature Range
Nonoperating	-65 to +165 degrees F
Ready, nonoperating	0 to +165 degrees F
Ready and operating	0 to +160 degrees F

3.3.5 Pressure. The oscillograph shall be capable of withstanding a pressure of 2.7 psia.

3.3.6 Humidity. The oscillograph shall withstand exposure to 95 percent relative humidity at 110 degrees F for eight hours without degradation of the finish or functional capabilities.

3.3.7 Operating life. The oscillograph shall be capable of withstanding at least 200 operation cycles without maintenance and thereafter shall meet the performance requirements of this specification. An operating cycle shall be as specified in 3.6 of Specification MIL-O-23560(Wep).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Testing and inspection shall be as described in Specification MIL-O-23560(Wep), except as follows:

4.1.1 Environmental tests. The oscillograph shall be subjected to all of the following environmental tests and thereafter shall meet the product characteristics of this specification.

4.1.1.1 Vibration. The oscillograph shall be conditioned at the temperature indicated in 4.1.1.1.1 for a minimum period of one hour prior to conducting the resonance survey. Vibrate the unit for a total of 9 hours (3.3.1 and Tables III and VII, in accordance with the requirements of Procedure XII, Specification MIL-E-5272 and the vibration deviations of 4.1.1.1.1).

4.1.1.1.1 Vibration deviations. The following are deviations from Procedure XII, Specification MIL-E-5272. Individual resonance surveys shall be conducted with vibration applied along each of three mutually perpendicular axes. The resonant and cycling periods in each axis shall be in accordance with Table VII, with one hour at the appropriate low temperature (-45 degrees F), one hour at room temperature, and one hour at 140 degrees F (see Figure 2).

Table VII

Resonant and Cycling Schedule

No. of Resonances	Time at Resonance at Each Temperature (Minutes)*	Cycling Time at Each Temperature (Minutes)
0	0	180
1	27	153
2	54	126
3	81	99
4	108	72

*Nine minutes at each resonance at each temperature in each axis.

4.1.1.2 Shock. The unit shall be mounted to simulate service conditions (Figure 2) and then mounted onto a shock test table so that a minimum of 90 percent of the table input will be applied to the service-

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condition mounting of the unit under test. Subject the unit to the shocks listed in Table VIII (in the condition specified in Table IV) as specified in the Shock, Test, Procedure IV, Specification MIL-E-5272, except as follows: Shock magnitude and duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in Specification MIL-E-5272.

Table VIII

Shock Schedule

Peak Magnitude (g)	Duration (ms)	Impact Direction	Number of Pulses
300	0.5 to 1.5	Longitudinally Aft.	3
85	0.5 to 1.5	Any direction except Longitudinally Aft.	6
60	8.0 to 10	Any	9
45	20.0 to 24	Longitudinally Fwd.	3

4.1.1.3 Acceleration. The oscillograph shall be subjected to the acceleration magnitudes and directions required by Table V for three minutes at room temperature.

4.1.1.4 Temperature.

4.1.1.4.1 Temperature; storage. The oscillograph, preserved and packaged in accordance with Section 5, shall be placed in a temperature controlled chamber and subjected to the following temperature cycle:

(a) Heat to 140 +5 -0 degrees F in 2.5 ±0.5 hours.

(b) Maintain at 140 +5 -0 degrees for 4 hours minimum.

(c) Cool to -45 +0 -5 degrees F in 2.5 ±0.5 hours.

(d) Maintain at -45 +0 -5 degrees F for 4 hours minimum.

(e) Return to room temperature in 2.5 ±0.5 hours.

Repeat this schedule for minimum of three cycles.

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4.1.1.5 Pressure test. The oscillograph shall be subjected to a pressure of 2.7 psia. The oscillograph shall meet the performance requirements specified in 3.3 after the pressure test.

4.1.1.6 Humidity test. The oscillograph shall be placed in a suitable test chamber and be subjected to 95 percent relative humidity for a period of eight hours at a temperature of 110 degrees F.

4.1.2 Acceptance test.

4.1.2.1 Magnetic interference. The oscillograph shall be subjected to the following magnetic tests.

4.1.2.1.1 Calibration procedure.

4.1.2.1.1.1 The pulse generator (Figure 3) is provided for calibration of the test setup. Connect pulse generator and amplifier filter (Figure 4) as shown in Figure 5. Set reversing switch to OFF position and pulse generator to ON. On the pulse generator connect point C to point A and adjust R7 to obtain a pulse width of 100 ±10 milliseconds. Adjust R14 to obtain a pulse amplitude of 200 ±2 millivolts across the output taps. Connect point C to point B for remainder of calibration.

4.1.2.1.1.2 Energize amplifier filter circuit and allow at least 15 seconds for circuit to reach full operating sensitivity. Connect oscilloscope to output of amplifier filter.

4.1.2.1.1.3 Set reversing switch to NORMAL and observe scope indication. After each pulse, set reversing switch to opposite position. Continue this procedure until peak amplitude has been determined. Record peak amplitude for reference.

4.1.2.1.2 Test procedure.

4.1.2.1.2.1 Mount oscillograph in pick-up coil in position shown in Figure 6. Mounting device must be of nonferrous construction; aluminum or wood are suitable materials.

4.1.2.1.2.2 Connect oscillograph to a suitable source of dc power, preferably a battery. If a power supply is used, care should be taken to physically isolate the supply from the test coil in order to prevent pick-up, such as of transients, generated within the supply.

4.1.2.1.2.3 Connect pick-up coil to amplifier filter and apply power to circuit. After 15 seconds warmup period, orient coil in room to minimize 60 cycle pickup. Fluorescent overhead lighting is undesirable.

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With coil situated for minimum pick up, the peak background level observed on the scope must be below the peak value observed in 4.1.2.1.1.3.

4.1.2.1.2.4 Install a loaded film magazine (40-foot roll of film) in the oscillograph to be tested. Energize oscillograph and observe magnetic disturbance level indicated on oscilloscope. The peak value of interference occurring during the time required to expose the entire film roll must be less than the peak value observed in 4.1.2.1.1.3 during the calibration.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-O-23560(Wep).

6. NOTES

6.1 Intended use. The oscillograph is intended for use in Torpedo Mk 46 Mod 0 (Exercise Head Mk 83 Mod 0).

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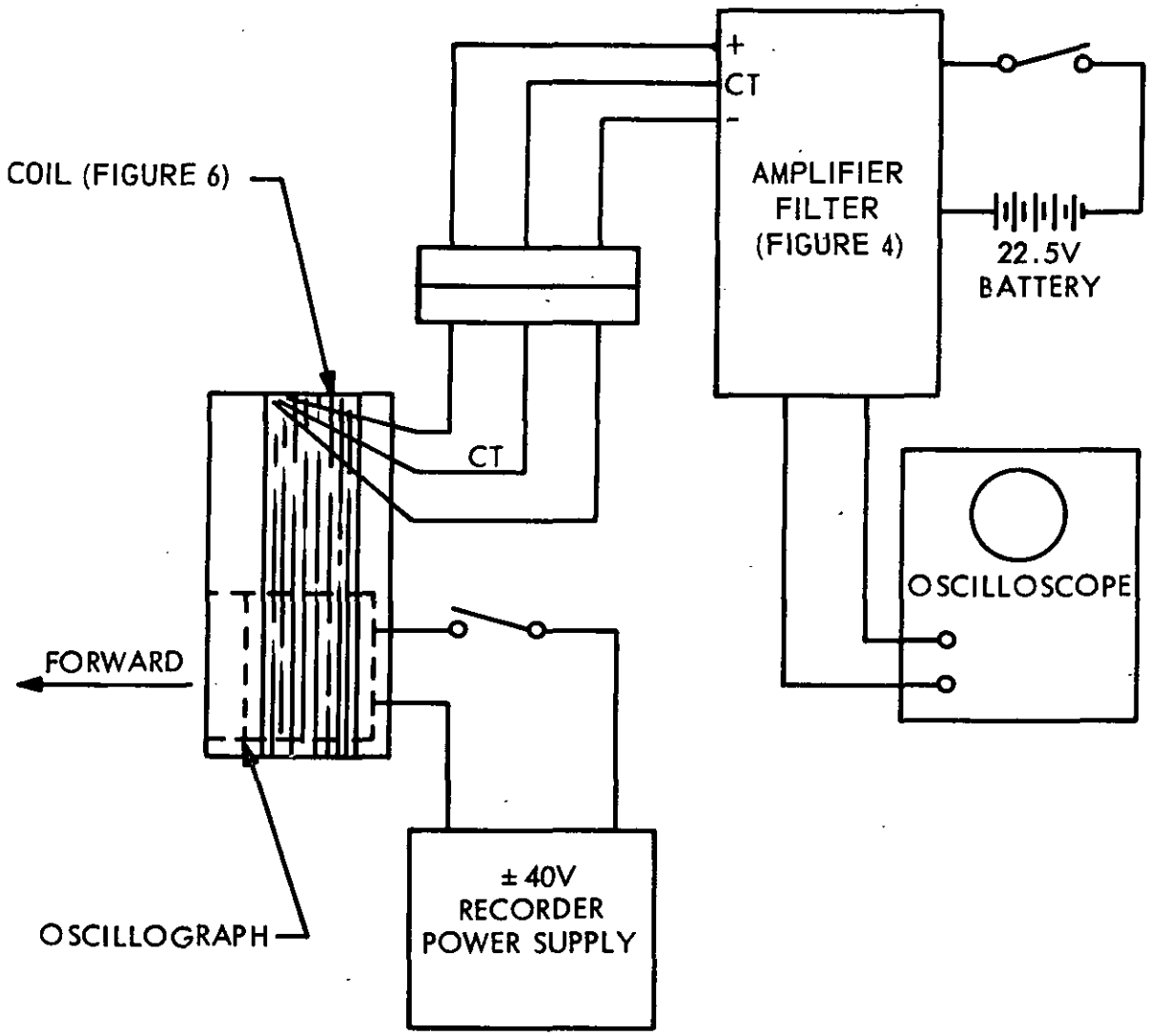


Figure 1
Test Setup for Measurement of Magnetic Interference

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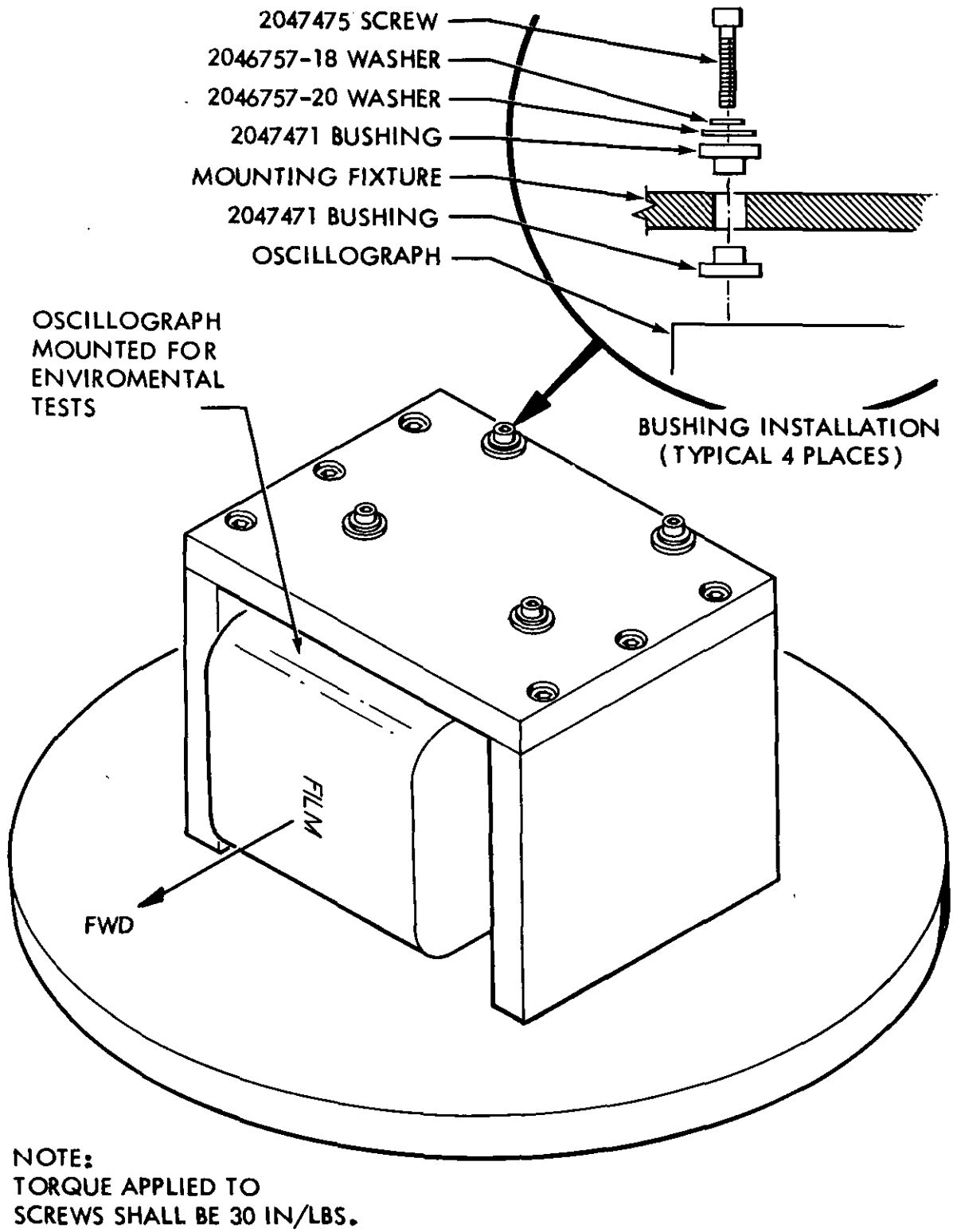
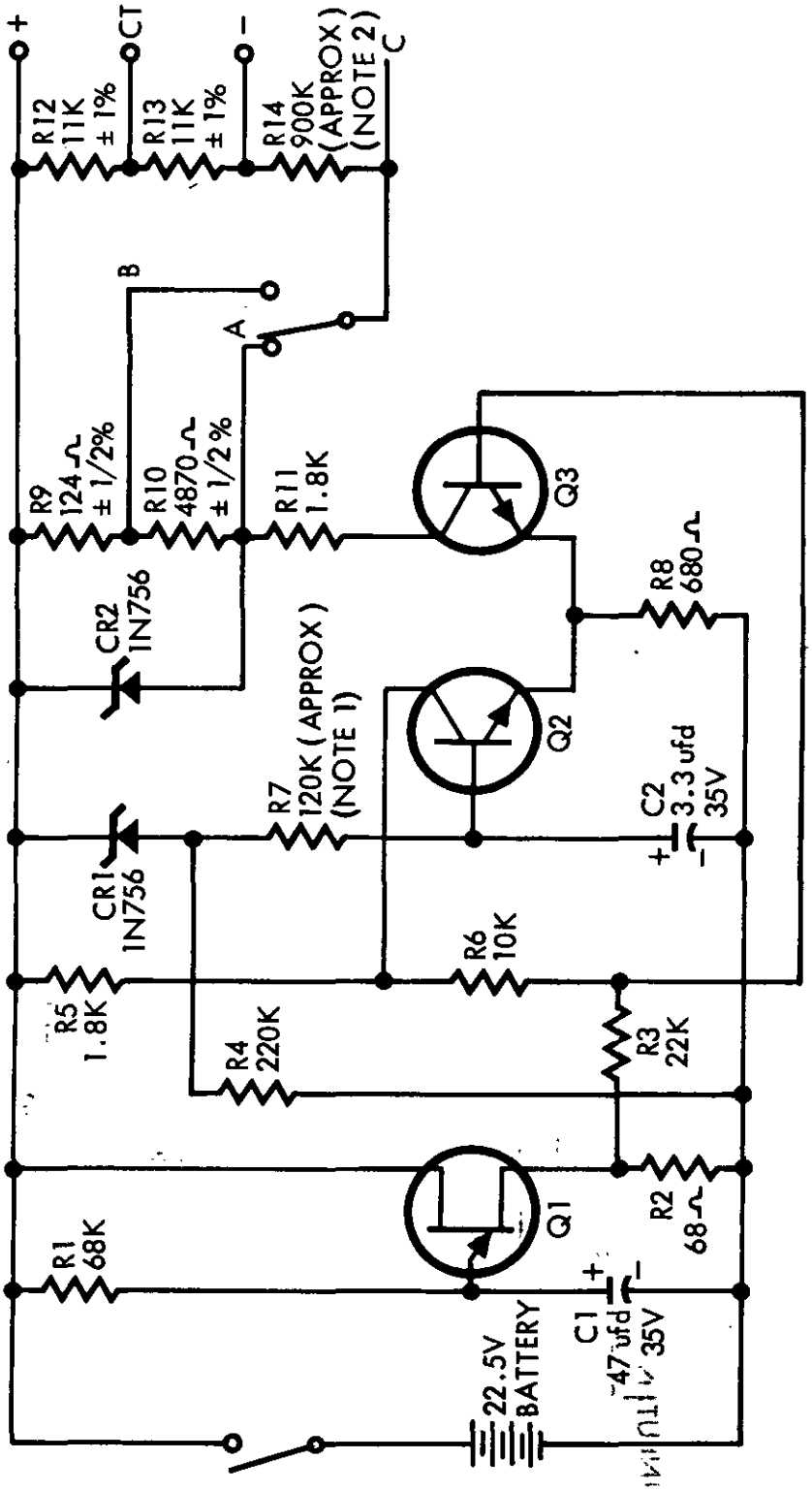


Figure 2
Simulated Service Mounting

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NOTE:

1. SELECT R7 TO PROVIDE 100 ± 10 MILLISECONDS PULSE WIDTH ACROSS CR2.
2. SELECT FOR 200 ± 2 MV ACROSS OUTPUT WHEN POINT C IS CONNECTED TO POINT A.
3. Q1 - 2N491, 2N492 OR 2N493
   Q2 AND Q3 - 2N336 OR 2N479
4. CAPACITORS - SPRAGUE 150D SERIES (OR EQUIV.)
5. RESISTORS - 1/2 WATT, 5%, UNLESS OTHERWISE SPECIFIED.

Figure 3
Pulse Generator for Calibration of Magnetic Interference Test Setup

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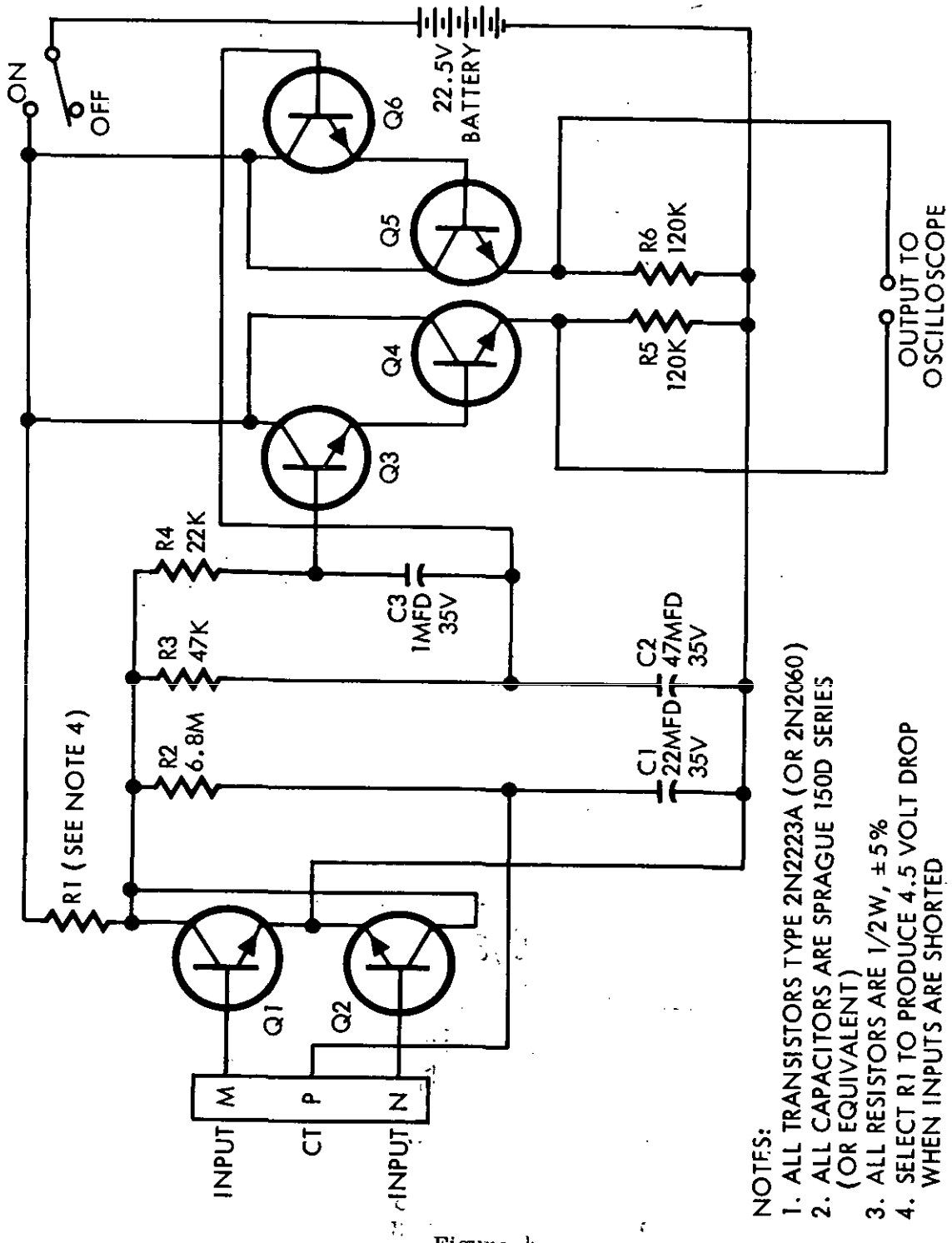


Figure 4
Filter and Amplifier for Measurement of Magnetic Interference

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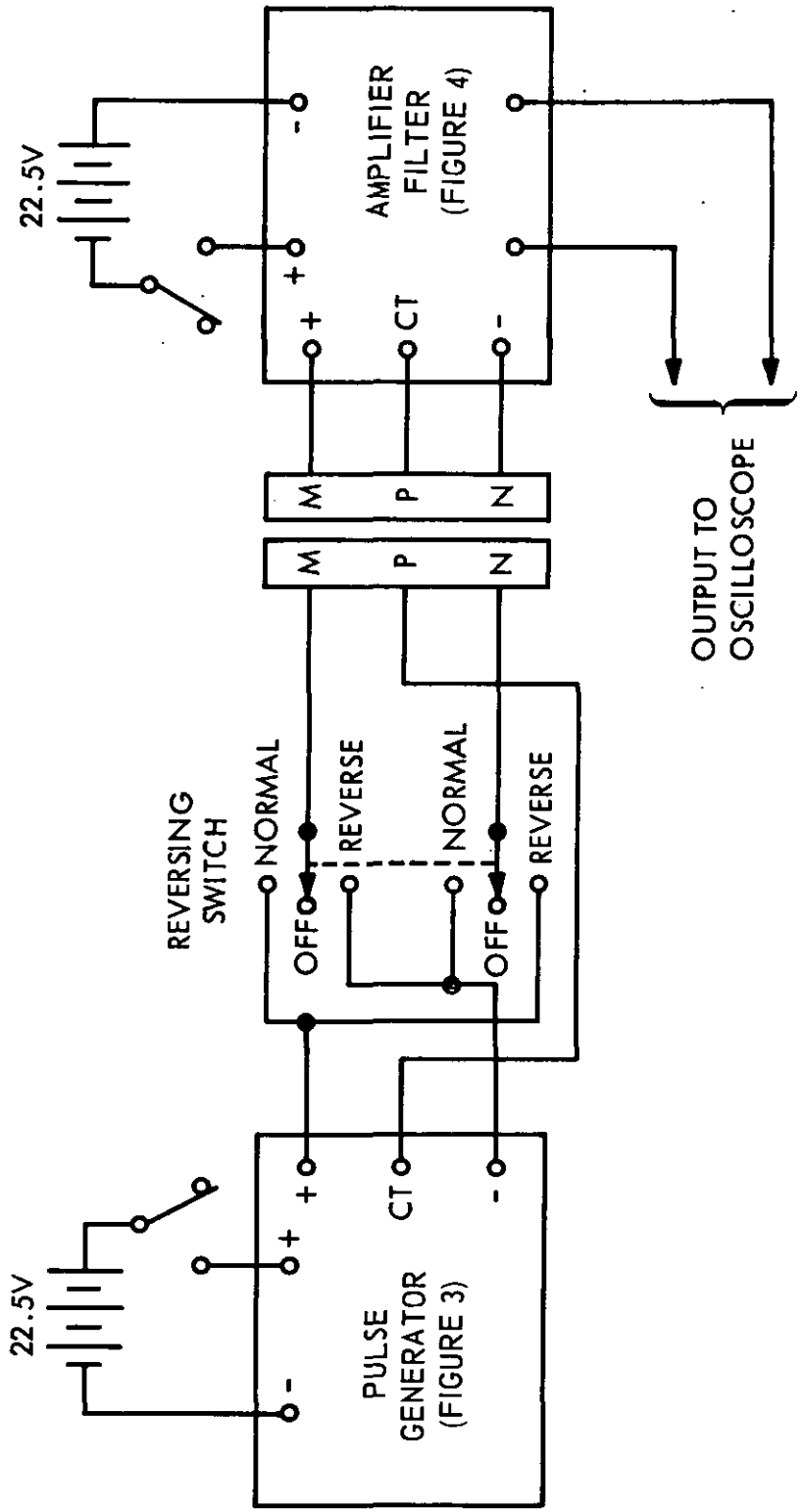


Figure 5
Calibration Test Setup

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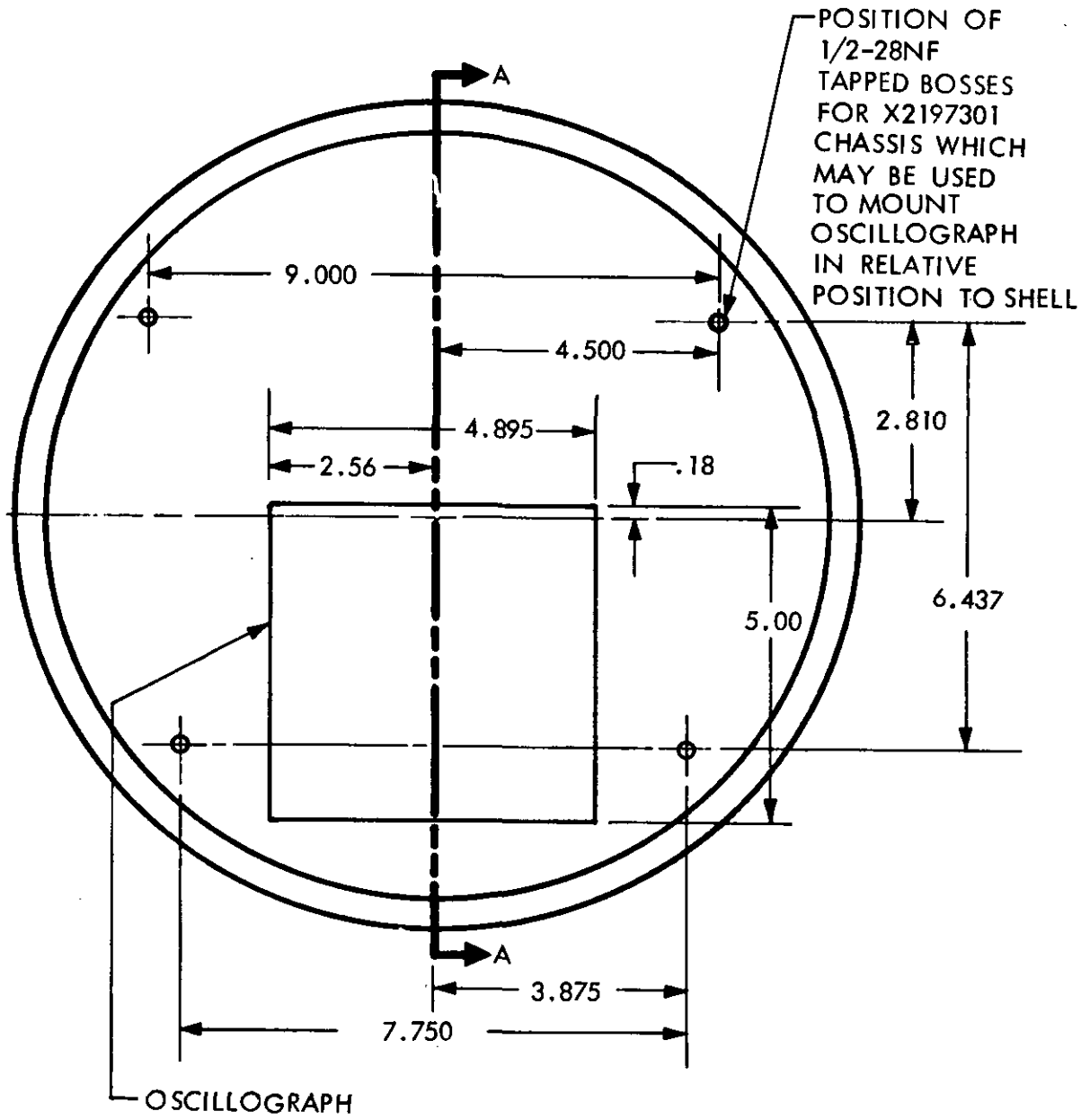


Figure 6 (Sheet 1)
Construction Details for Pick-up Coil Used in Magnetic Interference Measurements

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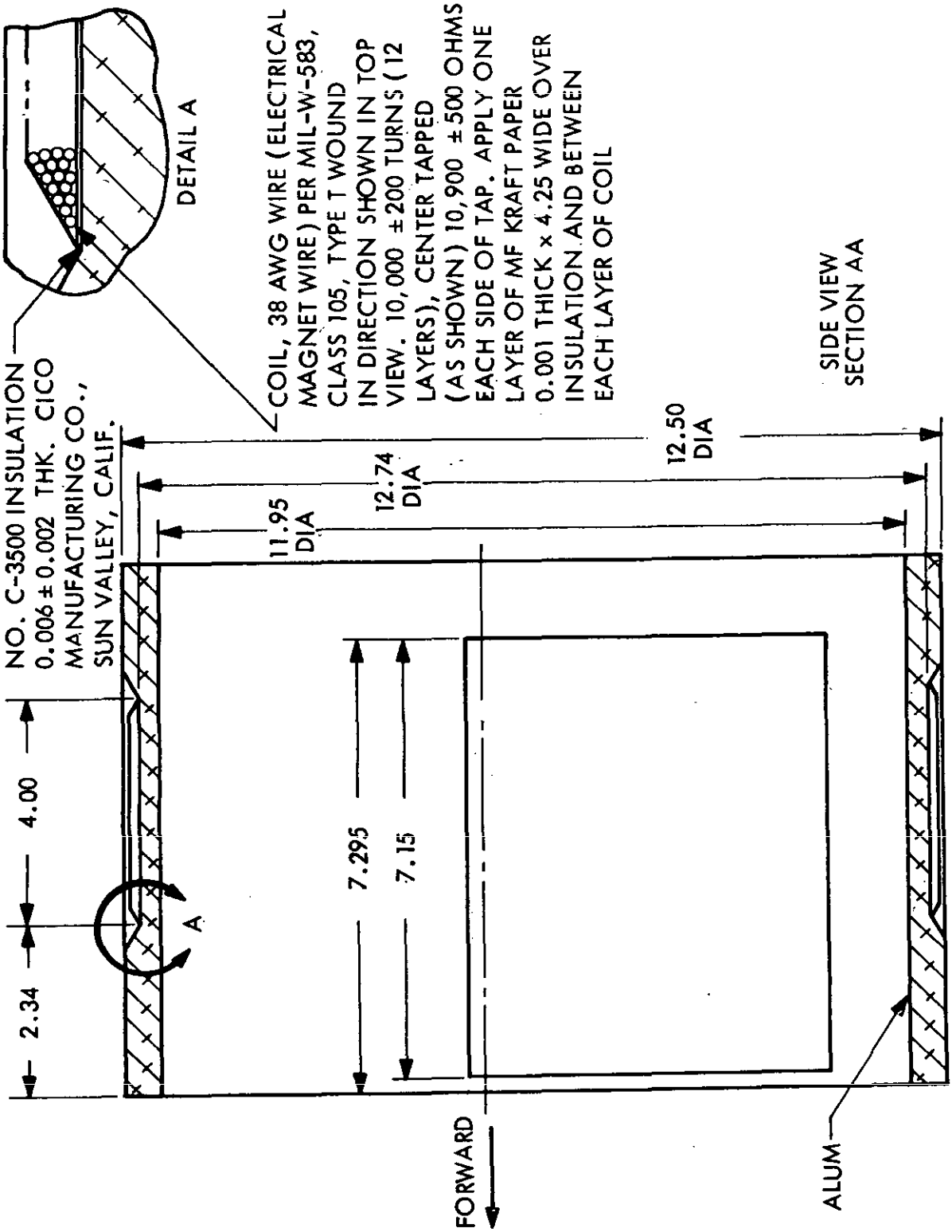


Figure 5 (Sheet 2)

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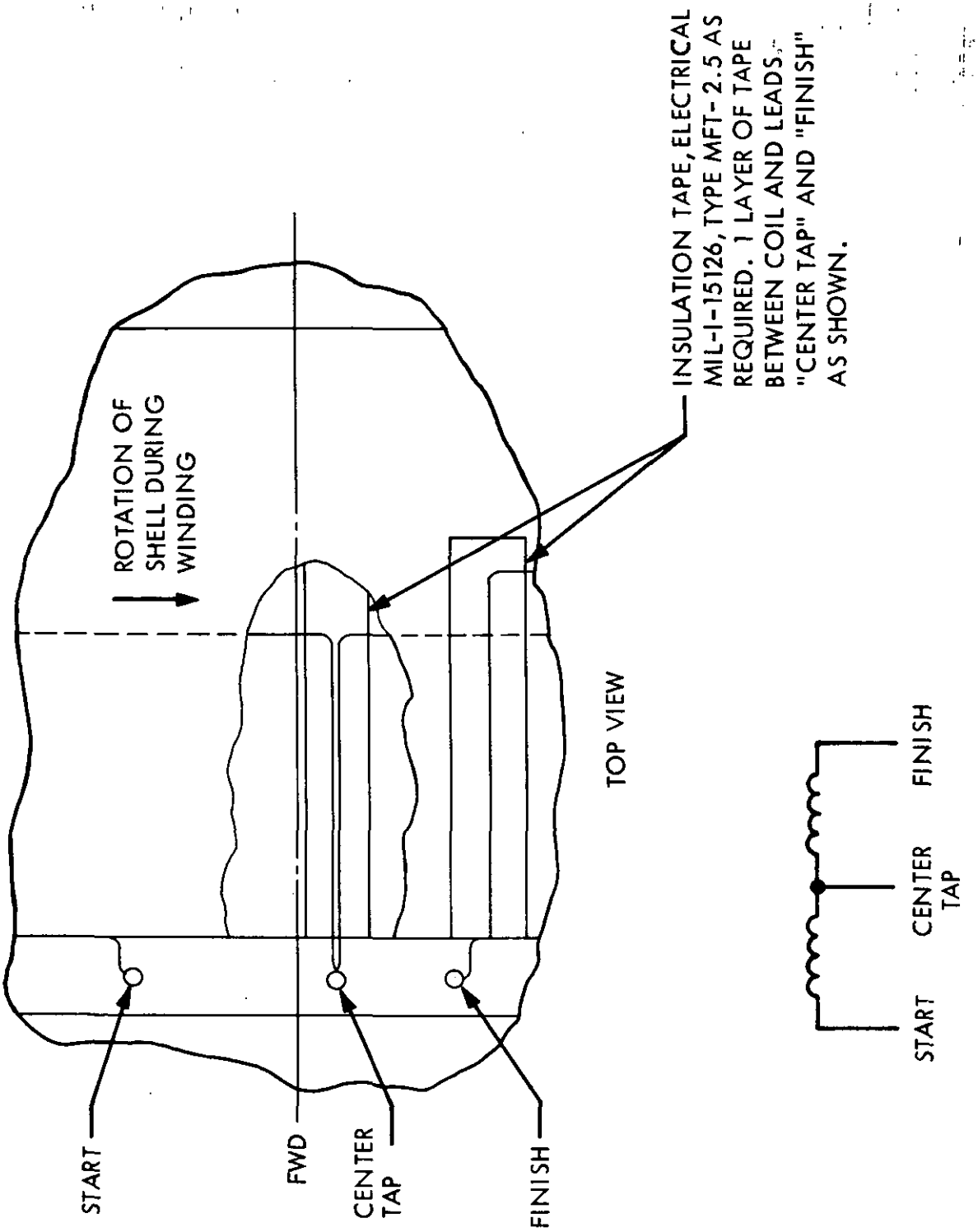


Figure 5 (Sheet 3)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-O-23560/5(Wep) Oscillograph, Type V

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Washington, D. C. 20360

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